

RUNHAN HUANG

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EDUCATION

Tsinghua University, Yao Class

September 2022 - Now

Bachelor of Science in Computer Science (GPA: 3.87 / 4.00)

Beijing

EXPERIENCE

MARS Lab, IIS & Shanghai Qizhi Institute

June 2024 – Now

Research Assistant

Beijing

- Advised by **Prof. Hang Zhao**, working on **Robot Locomotion** and **Robot Manipulation**

Du Lab, Kempner Institute, Harvard University

March 2025 – September 2025

Visiting Researcher

MA, USA

- Advised by **Prof. Yilun Du**, working on **Diffusion Planning**

PUBLICATIONS

- [P1] Zunzhe Zhang*, **Runhan Huang***, Yicheng Liu, Shaoting Zhu, Linzhan Mou, Hang Zhao. “Generative Control as Optimization: Time-Unconditional Flow Matching for Adaptive and Robust Robotic Control,” *ICLR LLA & ReALM-GEN Workshop*, 2026.
- [P2] Shaoting Zhu*, Baijun Ye*, Jiaxuan Wang†, Jiakang Chen†, Ziwu Zhuang, Linzhan Mou, **Runhan Huang**, Hang Zhao. “TTT-Parkour: Rapid Test-Time Training for Perceptive Robot Parkour,” *ArXiv*, 2026.
- [P3] **Runhan Huang**, Haldun Balim, Heng Yang, Yilun Du. “Flexible Locomotion Learning with Diffusion Model Predictive Control,” *ICRA*, 2026.
- [P4] **Runhan Huang***, Shaoting Zhu*, Yilun Du, Hang Zhao. “MoELoco: Mixture of Experts for Multitask Locomotion,” *IROS*, 2025.
- [P5] Shaoting Zhu*, Linzhan Mou*, Derun Li, Baijun Ye, **Runhan Huang**, Hang Zhao. “VR-Robo: A Real-to-Sim-to-Real Framework for Visual Robot Navigation and Locomotion,” *RA-L*, 2025.
- [P6] Shaoting Zhu, **Runhan Huang**, Linzhan Mou, Hang Zhao. “Robust Robot Walker: Learning Agile Locomotion over Tiny Traps,” *ICRA*, 2025.

RESEARCH EXPERIENCE

Generative Control as Optimization | *ArXiv 2026*

- Reinterpret flow-matching inference from fixed-schedule integration to iterative optimization by learning a time-unconditional, stationary velocity field
- Introduce an adaptive inference mechanism that dynamically allocates computation for contact-rich bottlenecks, alongside a training-free OOD anomaly detector using the vector field norm
- Scale the framework to π_0 -series VLA models and demonstrate superior sample efficiency and real-world robustness on a Galaxea R1 Lite humanoid robot for high-precision tasks

Flexible Locomotion Learning with Diffusion Model Predictive Control | *ICRA 2026*

- Reinterpret diffusion models as generative priors for MPC, using reward updates, constraint projection, and candidate ranking
- Introduce a reward-weighted denoising method to finetune the diffusion planner, improving locomotion performance
- Develop real-time diffusion planning techniques such as asynchronous execution and early caching; demonstrate zero-shot transfer on a Unitree Go2 quadruped

MoELoco: Mixture of Experts for Multitask Locomotion | *IROS 2025*

- Develop a multitask locomotion framework using a mixture-of-experts (MoE) architecture.
- Enhance sample efficiency and skill compositionality for reinforcement learning across multiple locomotion tasks.
- Project leader; contributed to model design, training infrastructure, evaluation and real-world deployment.

VR-Robo: A Real-to-Sim-to-Real Framework for Visual Robot Navigation and Locomotion | RA-L 2025

- Build a digital-twin simulation pipeline using 3D Gaussian Splatting for photo-realistic rendering.
- Propose an RGB-based sim-to-real transfer policy framework for navigation and locomotion.
- Participate in low-level locomotion training and real world experiment

Robust Robot Walker: Learning Agile Locomotion over Tiny Traps | ICRA 2025

- Design a proprioception-only locomotion policy that generalizes to terrains with small obstacles ("tiny traps").
- Introduce a two-stage reinforcement learning training framework and a novel benchmark for tiny-trap traversal.
- Co-lead benchmark design, policy training, evaluation and real-robot deployment.

PROJECTS

Automatic offside detection system | Computer Vision Course Project

- Design an automatic soccer offside detection pipeline.
- Construct a high-quality soccer offside dataset
- Develop an automatic soccer offside detection system by training deep learning models and applying computer vision methods

Implicitly Exploit the Modularity in Multimodal Large Language Models | Multimodal Machine Learning Course Project

- Propose a new gradient update strategy for visual instruction tuning
- Train a LVLM based on phi-2 model, achieving good performance

Humanoid Perceptive Locomotion | Intelligent System and Robotics Course Project

- Construct a two-stage teacher–student locomotion learning paradigm with PPO and DAgger, enabling a humanoid robot to traverse challenging terrain
- Demonstrate robust stair, slope, and gap traversal in simulation using only depth image and proprioception

Long Horizon Mobile Manipulation with Curiosity Enhanced Reinforcement Learning | Embodied AI Course Project

- Build a long-horizon mobile manipulation RL pipeline in ManiSkill, coupling navigation and articulated-object manipulation with a clean evaluation workflow
- Introduce a hash-based curiosity reward to mitigate local minima and drive novelty-seeking exploration across task stages
- Apply Leaky PPO to improve stability and sample efficiency, further boosting learning performance

Multi-agent Navigation with Vision Language Model | Practical Research Training Project

- Build a city-scale multi-agent navigation dataset combining Google Street View, autonomous driving data, and 3D Gaussian Splatting reconstructions, enriched with landmark, orientation, and action annotations to directly supervise spatial reasoning skills.
- Design a dual benchmarking protocol spanning static VQA-style prediction and 3DGS-based simulation, providing a comprehensive assessment of perception, planning, and coordination
- Benchmark the state-of-the-art VLMs, showing that current models fall significantly short, motivating new research on embodied multi-agent intelligence

SKILLS

Programming: C, C++, Python, Pytorch, ROS, ROS2

Robotics related software/hardware experience: isaacgym, Unitree A1, Unitree Go2, Galaxea R1 Lite Mobile Manipulator, PiperX Robot Arm

Key Course Taken: C/C++ Programming (A+), Calculus (A), Linear Algebra (A), General Physics (A+), Algebra and Computation (A+), Algorithm Design, Probability and Statistic, Machine Learning, Deep Learning, Multimodal Machine Learning (A+), Computer Vision (A), Theoretical Computer Science, Reinforcement Learning, Autonomous Driving (A+), Advanced Computer Graphics (A), Practical Research Training (A)

EXTRACURRICULAR EXPERIENCE

IIIS Soccer Team

Captain

- Led a soccer team and managed the soccer club of IIIS.

Jan 2024 – Feb 2025

Beijing